

GRAPHIC ERA HILL UNIVERSITY, DEHRADUN

SEMESTER VII

Name of Department: - Computer Science and Engineering

1. Course Code: **TCS762** Course Title: **NoSQL Database**
2. Contact Hours: L: **3** T: **0** P: **0**
3. Examination Duration (Hrs): Theory **3** Practical **0**
4. Relative Weight: CIE **25** MSE **25** ESE **50**
5. Credits: **3**
6. Semester: **VII**
7. Category of Course: **DSC**
8. Pre-requisite: Introduction to Big Data (TCS462), Big Data Analytics: Tools and Techniques (TCS 663)

9. Course Outcome:	After completion of the course the students will be able to: CO1: Demonstrate an understanding of the fundamental concepts, evolution, and aggregate-oriented data models of NoSQL databases, including Key-Value, Document, Column-Family, and Graph databases. CO2: Analyze distribution, replication, consistency, and scalability mechanisms used in NoSQL databases. CO3: Apply MapReduce and key-value database concepts for processing and managing large-scale data. CO4: Apply document database concepts to model, store, query, and manage semi-structured data for real-world applications. CO5: Evaluate Graph database concepts and their suitability for applications involving connected data, routing, location-based services, and recommendation systems.
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10. **Details of the Course:**

Sl. No.	Contents	Contact Hours
1	Unit 1: Why NoSQL? The Value of Relational Databases, Getting at Persistent Data, Concurrency, Integration, A (Mostly) Standard Model, Impedance	10

	Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL. Aggregate Data Models: Aggregates, Example of Relations and Aggregates, Consequences of Aggregate Orientation, Key-Value and Document Data Models, Column-Family Stores, Summarizing Aggregate-Oriented Databases. More Details on Data Models: Relationships, Graph Databases, Schemaless Databases, Materialized Views, Modeling for Data Access	
2	Unit 2: Distribution Models: Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication, Consistency, Update Consistency, Read Consistency, Relaxing Consistency, The CAP Theorem, Relaxing Durability, Quorums. Version Stamps, Business and System Transactions, Version Stamps on Multiple Nodes.	8
3	Unit 3: Map-Reduce: Basic Map-Reduce, Partitioning and Combining, Composing Map-Reduce Calculations, A Two-Stage Map-Reduce Example, Incremental Map-Reduce. Key-Value Databases: What Is a Key-Value Store, Key-Value Store Features, Consistency, Transactions, Query Features, Structure of Data, Scaling, Suitable Use Cases, Storing Session Information, User Profiles, Preference, Shopping Cart Data, When Not to Use, Relationships among Data, Multi-operation Transactions, Query by Data, Operations by Sets	9
4	Unit 4: Document Databases: What Is a Document Database?, Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Web Analytics or Real-Time Analytics, E-Commerce Applications, When Not to Use, Complex Transactions Spanning	9

	Different Operations, Queries against Varying Aggregate Structures.	
5	Unit 5: Graph Databases: What Is a Graph Database?, Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Connected Data, Routing, Dispatch, and Location-Based Services, Recommendation Engines, When Not to Use	9
	Total	45

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Sadalage, P. & Fowler	NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence	1 st	Pearson/Addison-Wesley, American	2012

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Dan Sullivan	NoSQL For Mere Mortals	1 st	Pearson Education India	2015
Dan McCreary and Ann Kelly	Making Sense of NoSQL: A guide for Managers and the Rest of us	1 st	Dreamtech Press, New Dehli	2013
Kristina Chodorow	Mongodb: The Definitive	2 nd	O'Reilly, United Kingdom	2013

	Guide- Powerful and Scalable Data Storage			
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2025-26 onwards